Liangyu (Andrew) Ding

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Education

The Chinese University of Hong Kong, Shenzhen, Shenzhen, China — Sep 2022 - present

B.S. Data Science and Big Data Technology

- Selected Core Courses: Advanced Convex Optimization (Ph.D level), Advanced Stochastic Process (Ph.D level), Dynamic Programming and Stochastic Control (Ph.D level), Stochastic Simulation, Numerical Methods, Deep Learning, C/C++ Programming, Data Structures, PDEs, Statistical Inference, Statistical Modeling in Financial Markets

Research Interest

Pricing and revenue management, AI for operations research, operations research for AI, data-driven decision making.

Publications and Preprints

[1] **Liangyu Ding**, Chenghan Wu, Guokai Li, Zizhuo Wang. Self-Improving Neural Pruning: A Graph Neural Network Framework for Scalable Mixed Bundle Pricing. *Under review*, 2026.

Research Projects

Self-Improving Neural Pruning for Scalable Mixed Bundle Pricing, — Apr 2025 - Jun 2026

Undergraduate researcher, *advised by Prof. Zizhuo Wang*

- Designed a GNN-guided pruning-then-optimization framework for mixed bundle pricing by encoding each instance as a compact customer-product graph and predicting segment-product assignment probabilities.
- Developed fixed and progressive cutoff pruning, GNN-guided local search, and iterative self-improvement to scale to larger non-additive instances without exact large-scale labels, recovering over 98% of optimal profit on small instances and outperforming bundle-size pricing on larger instances.
- Established a theoretical expressiveness result showing that, under mild distinguishability conditions, edge-output GNNs can recover the optimal product-assignment mapping.

GMRES with Householder Reflections, — May 2025 - Nov 2025

Undergraduate researcher, *advised by Prof. Yin Zhang*

- Characterized complete stagnation of restarted GMRES through equivalent Hessenberg and residual-orthogonality conditions for non-symmetric linear systems.
- Proposed UGMRES, a Householder-reflection restart strategy that reorients the Krylov starting vector with low per-iteration overhead and breaks stagnation on constructed pathological systems against GMRES-DR, GCRO-DR, and GMRES-SDR benchmarks.

Industry Experience

SOOCHOW Securities, Shanghai, China — May 2024 - Sep 2024

Quantitative Research Intern

- Developed an enhanced CSI 300 Index fund selection system, generating trading signals by using a composite of a 2-month Momentum factor and Tracking Deviation Excess Return, and selecting funds using LightGBM for return prediction, achieving an annualized return of 8.92%.

Course Projects

Image Deblurring

Ranked 1^{st} out of 101 students in the Kaggle Competition

CUHK(SZ) *Numerical Methods* Final Project

- Developed and implemented multiple image deblurring algorithms to improve numerical stability and deblurring quality, addressing challenges posed by singular matrices.

Solving Lasso via ADMM and Proximal-Type Algorithms

CUHK(SZ) *Advanced Convex Optimization* Final Project

- An implementation of ADMM, Proximal Gradient Method and Fast Iterative Shrinkage-Thresholding (FISTA) to solve LASSO problems.

Solving Semidefinite Programs: Primal-Dual Interior-Point Method
CUHK(SZ) *Advanced Convex Optimization* Project

- An implementation of predictor-corrector primal-dual interior-point method in Zhang, (1998) to solve standard form SDP.

Computing Wasserstein Barycenter via Linear Programming
CUHK(SZ) *Optimization in DS and ML* Final Project

- An implementation of predictor-corrector interior point method with the single low-rank regularization method in Ge et al. (2019) for the Pre-specified Support Barycenter Problem, MAAIPM in Ge et al. (2019) and Algorithm 3 in Cuturi and Doucet, (2014) for the Free Support Problem.

Presentations

Self-Improving Neural Pruning: A Graph Neural Network Framework for Scalable Mixed Bundle Pricing (previously entitled Learning to Price Bundles: A GNN Approach for Mixed Bundling)

- POMS-HK Meeting, Shenzhen, China, Jan 2026.

Honors and Awards

Dean's List, The Chinese University of Hong Kong, Shenzhen 2023-2025
Undergraduate Research Award, The Chinese University of Hong Kong, Shenzhen 2025

Skills

Languages: English (Fluent, TOEFL 103, GRE 332), Mandarin (Native)
Computer Languages: Python, R, Matlab, C++, Gurobi, COPT, LaTeX